

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 316 (January 2024)

Power Electronics Laboratory

Final Project Report

Section: A2

Group: 04

**Frequency Stabilizer
(Variable Frequency AC to Desired Fixed
Frequency)**

Course Instructors:

Asikur Rahman Jowel

Lecturer, Department of EEE, BUET

Swmic Majumder

Department of EEE, BUET

Signature of Instructor: _____

Academic Honesty Statement:

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

Signature: _____ Full Name: Mahadi Zaman Student ID: 2006044	Signature: _____ Full Name: Shamiul Islam Anik Student ID: 2006045
Signature: _____ Full Name: Ahamad Abtahi Student ID: 2006051	Signature: _____ Full Name: Md Nazmus Shakib Tushar Student ID: 2006052
Signature: _____ Full Name: Md Maksudur Rahman Turzo Student ID: 2006053	Signature: _____ Full Name: Joyonta Debnath Student ID: 2006054
Signature: _____ Full Name: Anas Rohan Student ID: 2006064	

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1 Abstract

This project, titled "*Frequency Stabilizer (Variable Frequency AC to Desired Fixed Frequency)*," focuses on stabilizing variable frequency alternating current (AC) to deliver a consistent and desired fixed frequency AC output. The system follows a structured process consisting of four main stages: rectification, inversion, filtering, and control.

The input to the system is variable frequency AC, which is first rectified to convert it into direct current (DC). The rectified DC signal is then inverted back into AC at the desired fixed frequency using an inverter circuit. A frequency and duty cycle controller governs this process, ensuring accurate control over the output frequency by regulating the inverter's switching parameters. Finally, an AC filter is applied to smooth the output waveform, resulting in stabilized frequency AC.

The proposed frequency stabilizer is highly effective in applications requiring consistent AC frequency output, regardless of input fluctuations. It is suitable for scenarios where equipment or systems depend on precise and stable AC supply, such as power systems, industrial machinery, and sensitive electronic devices. This system offers an efficient and reliable solution for converting variable AC into a fixed, stable frequency with minimal distortion.

2 Introduction

The demand for consistent and stable AC frequency is critical across various applications, including industrial machinery, power systems, and sensitive electronic equipment. However, power sources or grid systems often produce variable frequency AC due to fluctuations, load changes, or inconsistent generation processes. These variations can disrupt operations, reduce efficiency, and cause damage to systems that rely on a fixed frequency AC supply. To address this challenge, the project titled "*Frequency Stabilizer (Variable Frequency AC to Desired Fixed Frequency)*" is designed to ensure the delivery of a consistent and stable AC frequency regardless of input fluctuations.

The proposed system works by converting the variable frequency AC into a stabilized fixed frequency output through a systematic process involving rectification, inversion, and filtering, controlled by a frequency and duty cycle controller. The rectification stage converts the variable AC input into direct current (DC), which eliminates frequency variations. The DC signal is then fed into an inverter to generate AC at the desired frequency. To ensure the output waveform is smooth and free of distortions, an AC filter is incorporated in the final stage. A dedicated controller manages the inverter switching frequency and duty cycle, ensuring precise control of the output frequency.

This frequency stabilizer is a versatile solution for modern energy systems and power electronics, where frequency stability is paramount. It can be employed in renewable energy systems, such as wind or solar power, where output frequency often fluctuates, or in industries requiring precise AC frequency for optimal performance. The project not only enhances reliability but also improves the efficiency of AC-powered systems by delivering a stable and desired frequency output.

3 Design

3.1 Problem Formulation

Maintaining a stable frequency is crucial for the proper operation of electronic machines and power systems. Stable frequency ensures seamless energy transfer in electrical machines, such as motors and transformers, operating at optimal performance levels. Reduces inefficiencies caused by reactive power and frequency drift, thus conserving energy. Precise synchronization avoids phase mismatches, ensuring smooth operation in applications like motor drives, conveyor belts, and communication systems. Minimizes mechanical and electrical stresses on components caused by frequency fluctuations. Prolongs the lifespan of devices by reducing wear and tear due to erratic operations or overheating. Stable frequency (e.g., 50 Hz or 60 Hz) ensures proper operation of large-scale electrical networks and synchronization of generators.

On the other hand Frequency instability can disrupt the functioning of sensitive devices such as microcontrollers, digital clocks, and communication systems. In power systems, it can lead to synchronization loss, causing malfunctions in motors and generators. Components like capacitors, inductors, and transformers can experience irregular loading, leading to accelerated aging and early failure. Fluctuating frequencies introduce harmonics and noise, which generate additional heat in components like resistors and transformers. These inefficiencies lead to wasted energy and higher operational costs.

3.1.1 Identification of Scope

The following are the Primary objectives of the project:

- Develop a robust circuit capable of stabilizing an unstable AC signal by compensating for frequency variations and maintaining consistent output.
- Integrate a user-friendly control mechanism (e.g., potentiometers) to allow precise adjustment of the desired output AC frequency within a predefined range.
- Incorporate a digital display system, such as an LCD or LED screen, to provide real-time feedback of the output AC frequency, ensuring users can monitor performance effectively.
- Optimize the system to minimize error and drift, ensuring the output frequency remains highly accurate and stable under varying load or input conditions.
- Design the system to support a range of frequencies, making it adaptable for various applications and requirements.
- Ensure the circuit operates with high energy efficiency to reduce power losses and improve sustainability.
- Create a modular design that can be scaled up or down for different power and frequency stabilization requirements.

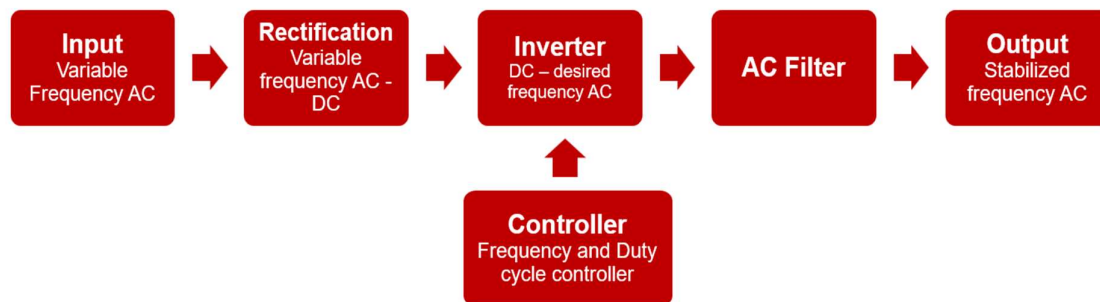
3.1.2 Literature Review

In order to maintain constant AC frequencies, early frequency stabilization systems used mechanical stabilizers, such as revolving machinery. These techniques were frequently ineffective and sluggish to react, which made it difficult to maintain grid stability.

Frequency stabilization has undergone a revolution with the introduction of power electronics. In order to accomplish quick and accurate frequency control, modern systems use components like inverters and converters in conjunction with digital control methods. The effectiveness and responsiveness of frequency stabilization techniques have been greatly enhanced by these developments. Despite advancements in technology, problems still exist. Conventional approaches can become unstable due to changes in input frequency and load, especially in power systems that are controlled by power electronics.

Devices like inverters and converters are essential for integrating distributed energy resources into modern power systems, enhancing reliability and power quality. An increasing trend is the integration of frequency stabilization technologies with the smart grids. This convergence makes adaptive control and real-time monitoring possible, enabling more effective and responsive frequency management.

3.2 Design Method



- Alternating current (AC) systems, frequency (f) and harmonic distortion can cause instability or fluctuations. Direct current (DC), however, is characterized by a constant value with no inherent frequency component.
- To overcome frequency fluctuations or remove harmonics in an AC system, the AC signal can be converted to DC. During this process, harmonics and frequency components are removed, leaving a steady DC voltage. This process is done by using a rectifier. We do not need a controlled rectifier as we shall control our output in the inverter unit.
- Then this DC signal can be reconverted into AC with the desired frequency and waveform characteristics. The resultant DC voltage can be used as an input to an inverter, which creates an AC signal with the desired frequency desired.
- An IC SG3524 is can be used as the controller of the inverter. It can provide adjustable frequency and duty cycle control for inverter operations.
- The frequency of the SG3524 is determined by an external timing resistor (Rt) and capacitor (Ct) connected to Pin 6 (RT) and Pin 7 (CT). Frequency for Rt = 55.3k and Ct = 0.47uF:

$$f_c(kHz) = \frac{1.3}{R_t \times C_t}$$

$$f_c(kHz) = \frac{1.3}{55.3 \times 0.47}$$

$$f_c(kHz) = 50 \text{ Hz}$$

- For a Push-pull inverter we need two alternating 180 degrees phase shifted pulse signals. We can get them from the 12 and 13 pin.
- Lastly, we may need a Filter to smooth out the inverter output. The push pull inverters square output can be turned into a sin output by using a RC third order filter. As our desired output frequency will be around 50Hz. So we will be taking a cutoff around at 70 Hz. The cut off frequency for $R = 5k$ and $C = 0.47\mu F$.

$$f_c(kHz) = \frac{1}{2\pi \times R \times C}$$

$$f_c(kHz) = \frac{1}{2\pi \times 5k \times 0.47\mu}$$

$$f_c(kHz) = 67.72 \text{ Hz}$$

3.3 Circuit Diagram

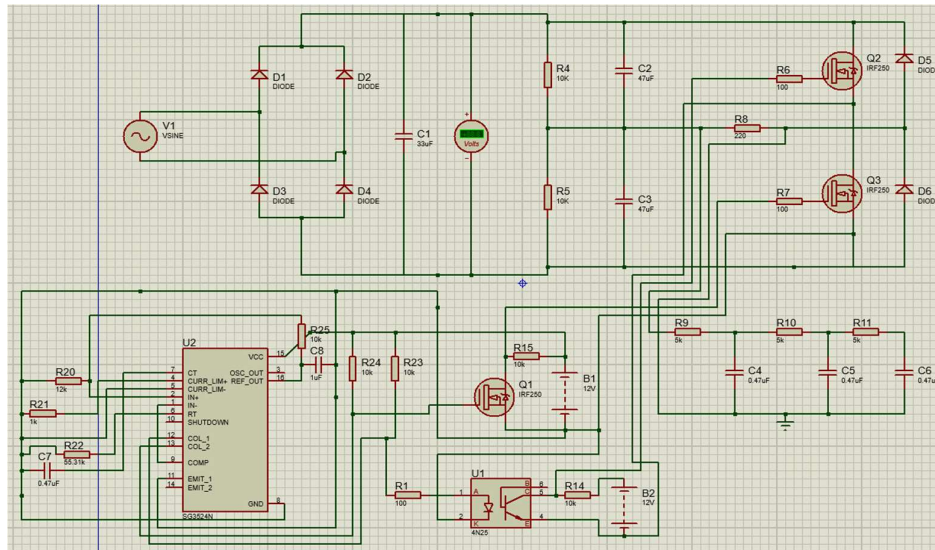


Fig: Circuit Diagram

3.4 Simulation Model

The output of the inverter in simulation:

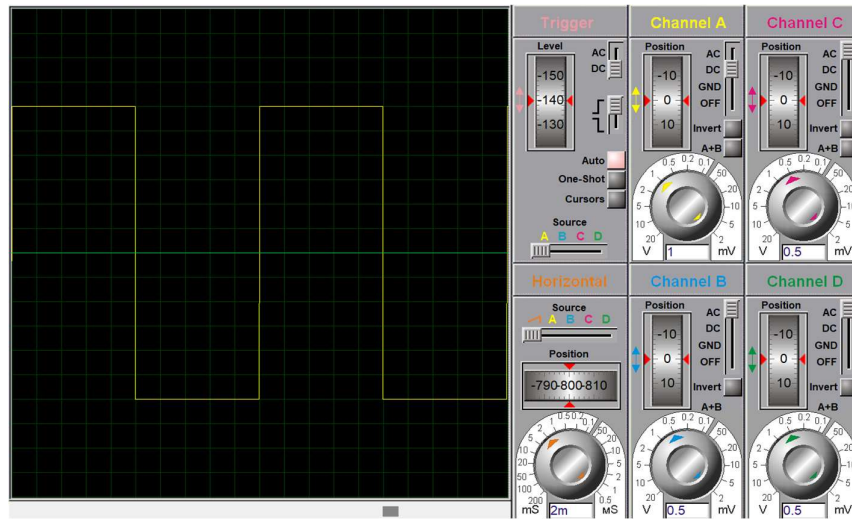


Fig: Output Square Wave of the inverter

The output of the inverter is a square wave with a peak-to-peak voltage equal to the DC voltage source. In the simulation we used a 12V DC source from the rectifier as the input of the inverter. So, the output is accurate. The RMS voltage of the output signal is half of the input DC voltage, as we used a half bridge inverter. Using a full bridge inverter would not have lost half of voltage.

We used a third order RC filter which had a cutoff frequency of about 65 Hz. We kept the cutoff frequency at 65 Hz because our desired output is normally 50 Hz and in special cases, we may need 60 Hz output. So, we used a pot of 5k and the capacitor value was 0.47 μ F.

The output of the Filters in simulation:

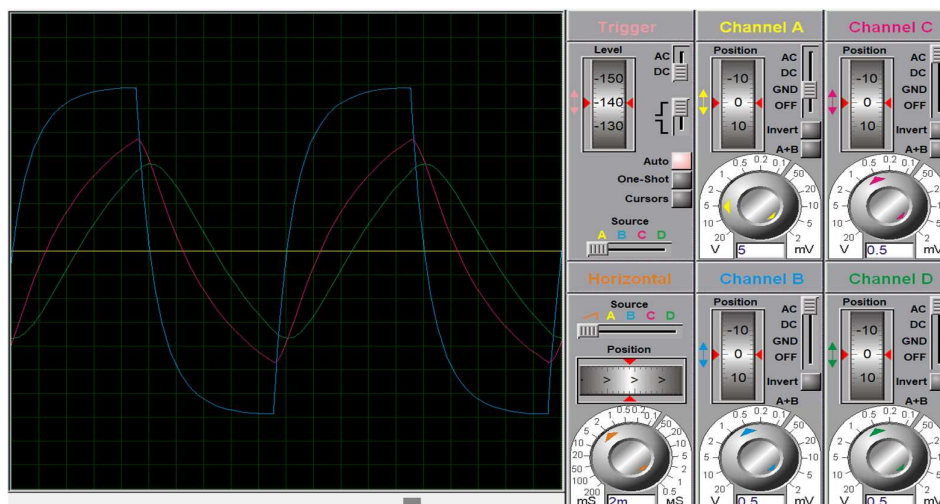


Fig: Output from the filter

This signal was fed into the smoothing filter to smooth out the edges. This process decreases the amplitude value but it turns the signal into a sinusoidal wave after three consecutive inversion operation. As we see the first filter turned the square wave signal into a Sawtooth. The output of the second filter is good enough for Realtime operation. But another filter was added to get the best value. In the plot the blue plot is the output of the First order filter. Red one is the output of the second order filter and the green one is the output of the third order filter, which is our desired output. The Filter we used is the optimum filter that gives the best result for minimum voltage drop. If we use other filter either the voltage drops increases or the filtration is not very good.

For RC filter with 1K resistor and 0.47uF capacitor, and 1k resistor and 4.7uF capacitor

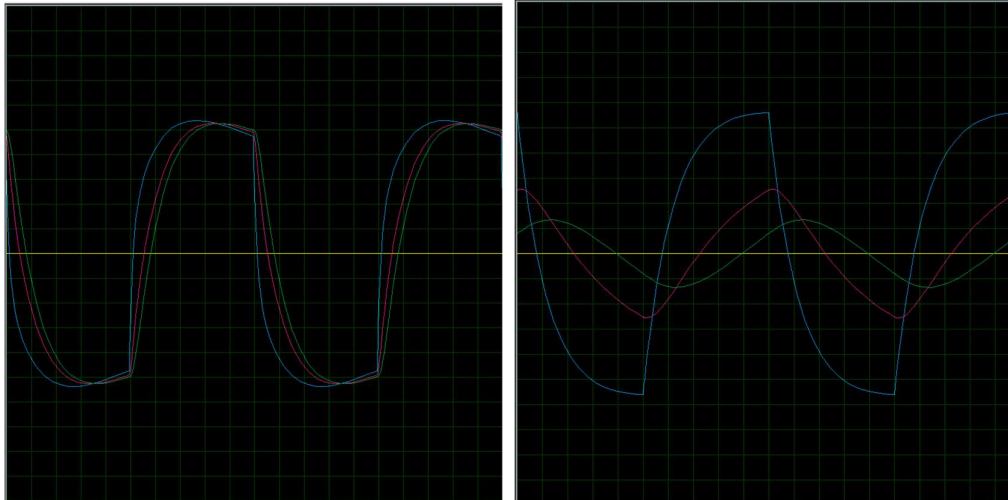


Fig: Output from filter with different parameter

3.5 Hardware Design

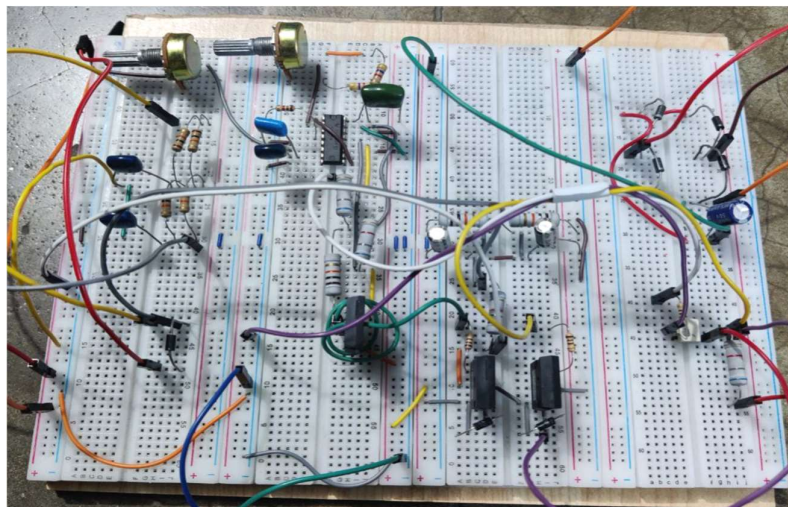


Fig : Hardware Design of frequency stabilizer .

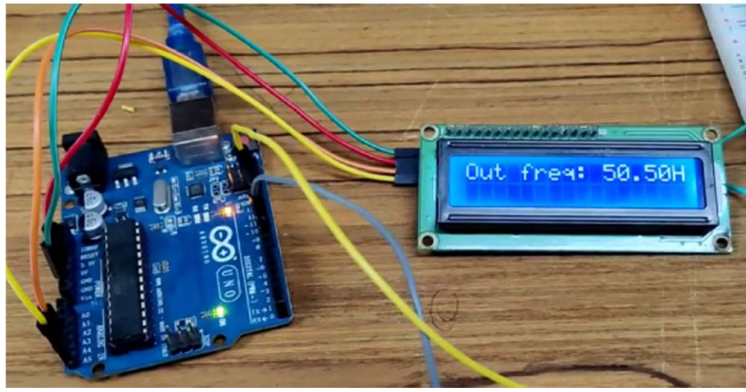


Fig: Output LCD Display for Realtime monitoring

4 Implementation

4.1 Description

In the inverter controller design, two potentiometers were used to enable manual adjustments:

1. Frequency Control: Adjusts the frequency of the inverter output.
2. Duty Cycle Control: Modifies the duty cycle of the PWM signal.

The design required three power input sources:

- Unstable Frequency AC Input: This serves as the primary input signal.
- DC Supply for the Controller: Powers the controller circuit.
- DC Supply for the Optocoupler: Provides power for signal isolation and safety.

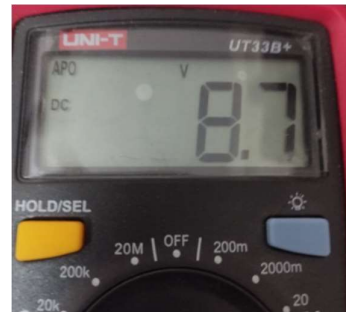
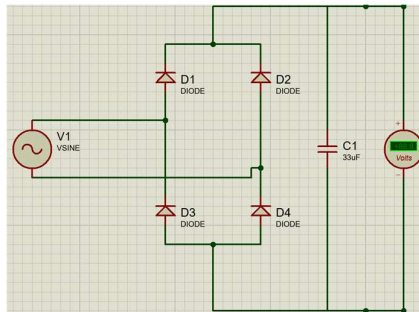
To simplify the system, the rectified DC voltage from the unstable AC input can be used to power the controller. This reduces the required input sources from three to two: AC Input and A single DC Source. Further optimization is possible by completely eliminating external DC sources. This can be achieved by isolating the grounds of the controller and optocoupler using internal circuit techniques, allowing the rectified DC voltage from the input AC signal to power both the controller and optocoupler.

An LCD display was implemented to monitor the inverter's performance in real-time. Key features include:

- Microcontroller: An Arduino Uno was used to process and display the output data.
- Input Signal: The Arduino received input directly from the inverter output.
- Signal Rectification: Since the inverter output was a rectangular wave spanning from a positive voltage to a negative voltage, it needed to be rectified before being fed into the Arduino. This ensured compatibility with the digital controller, which requires positive-only signals.
- The inverter output was a rectangular wave, easily interpretable as a digital signal after rectification. Rectification converted the bipolar output (positive to negative voltage) into a unipolar signal suitable for processing by the digital controller.

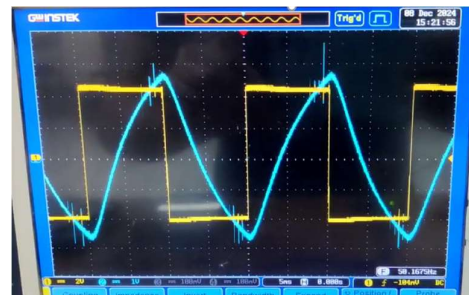
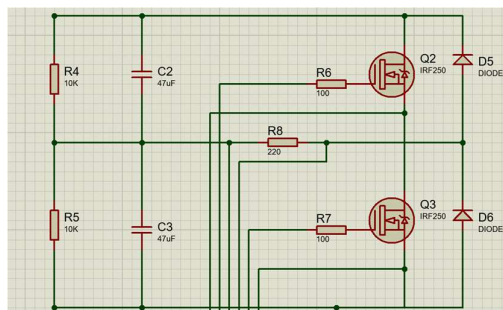
4.2 Experiment and Data Collection

1. Rectifier: Full-Bridge Non-Controlling Rectifier



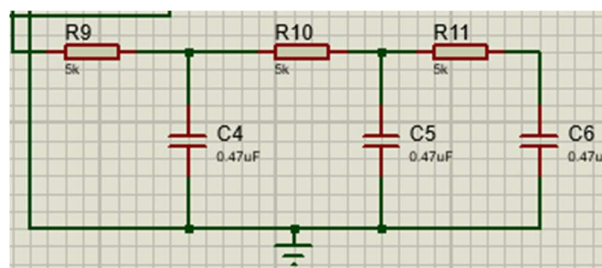
- Input Voltage: A 20V peak-to-peak sinusoidal signal.
- Input Frequency: Variable; manually adjusted using a function generator.
- Output Voltage: A steady 9V DC signal. A capacitor was included to smooth the output and eliminate ripples, providing a flat DC voltage.
- Output Frequency: 0 Hz, since the capacitor ensures no periodic fluctuations in the rectified signal.

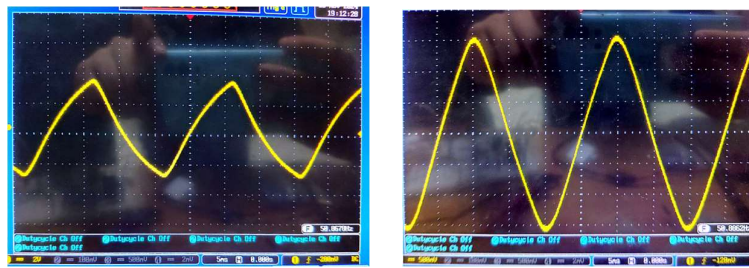
2. Inverter: Half-Bridge Inverter



- Input: 9V DC from the rectifier.
- Input Frequency: 0 HZ.
- Output: 8V peak-to-peak square wave signal.
- Output Frequency: 50 Hz.

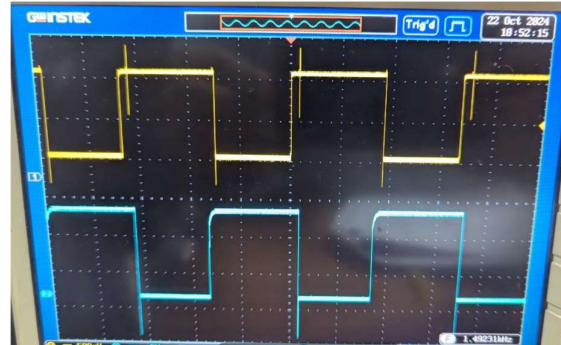
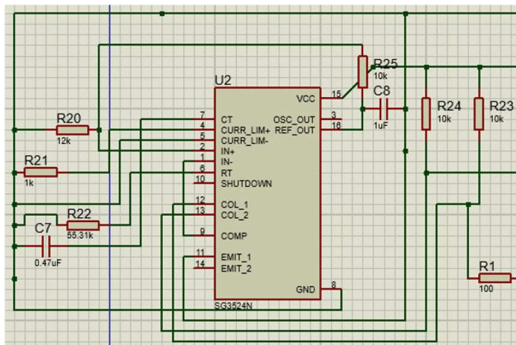
3. Filter: 3rd Order RC Low-Pass Filter





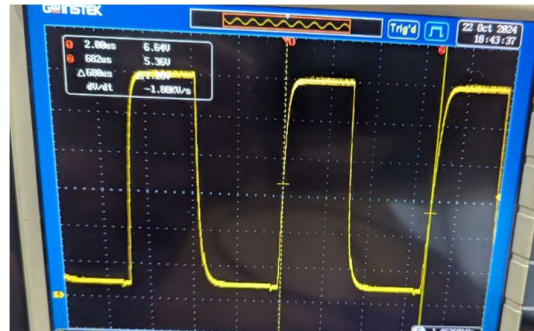
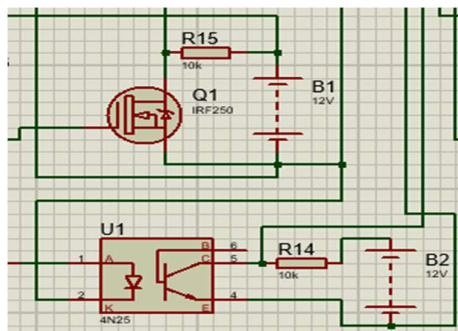
- Input: 8V peak-to-peak 50 Hz square wave signal from the inverter.
- Cutoff Frequency: 67 Hz. This cutoff was designed to allow the fundamental frequency of the signal to pass while attenuating higher-order harmonics.
- Output: 4V peak-to-peak sinusoidal signal.
- Output Frequency: 50 Hz.

4. Controller: SG3524 Pulse generator



- Input: 9V DC input.
- Output: Two alternating pulse rectangular signal.
- Output Frequency: 50 Hz. Controlled by the potentiometer.

5. Ground Isolator: An optocoupler was used.



- Power Input: 9v DC input.
- Reference Input: Pulse signal from Controller output pin 12.
- Output: A pulse rectangular signal. But with a different ground reference.
- Output Frequency: 50 Hz. Controlled by the potentiometer.

4.3 Data Analysis

Rectifier: For a full bridge diode rectifier:

- For input $V_{in}=20$ V peak-to-peak and diode voltage drop = 0.7V.
- $V_{out} = 10 - 2*0.7 = 8.6$ V peak .
- With a filter, $V_{out} = 8.6$ V DC $\approx$ 9V DC.
- Without a filter, $V_{out} \approx 6.37$ V DC.

Which is accurately equal to the experimental value.

Inverter: For a half bridge inverter:

- For $V_{in} = 8.6$ V
- $V_{out,p-p} = 8.6$ V
- $V_{out} = \pm V_{in}/2$ Volts
- $V_{out} = 4.3$ Volts

Which is equal to the experimental value 4V.

Filter: For a RC filter:

- For $V_{in} = 8.6$ V
- $V_{out,p-p} = 8.6$ V
- $V_{out} = \pm V_{in}/2$ Volts
- $V_{out} = 4.3$ Volts

Which is equal to the experimental value 4V .

$$\omega_c = 2\pi f_c = 2\pi \times 67 = 420.97$$

$$\omega = 2\pi f = 2\pi \times 50 = 314.15$$

$$\begin{aligned} |H(\omega)| &= \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^6}} \\ &= \frac{1}{\sqrt{1 + \left(\frac{314.15}{420.97}\right)^6}} \\ &= 0.92 \end{aligned}$$

For a 4 volt the output should be 3.68 V. In the Experimental value the output is 3 Volts so it's quite accurate.

4.4 Results

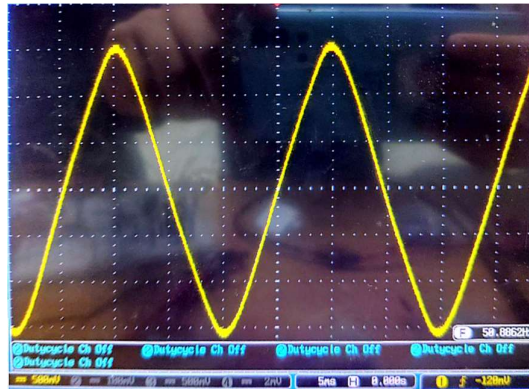


Fig: Output Sine wave of desired frequency.

From the output we can see that we got almost a sinusoidal waveshape of 50 Hz. Without RC filter it was a square waveform. By varying the frequency, we observed that the output frequency is always constant for a wide range of frequency input. The peak of the sinusoidal output is low, but it can be increased by using a transformer. By using a transformer if we increase the voltage output the power does not decrease because all the components that we used in the circuit have a high current rating so the circuit can provide a high amount of current.

We can also change the frequency of the output by changing the Potentiometer in the controller circuit. Also, we can provide a DC offset in the signal, although the DC offset will not be transformed in to a larger voltage if we use a transformer.

5 Design Analysis and Evaluation

5.1 Novelty

1. Integration of Modular Units

- The project seamlessly integrates key functional units: Rectification, Controller, Inverter, and Filtering stages, ensuring efficient conversion from unstable AC frequency to a stabilized desired frequency output.
- The use of modular components such as Full-Bridge Rectifier, IC3524 controller, Half-Bridge Inverter, and RC Multistage Filters makes the design flexible and scalable for various applications.

2. Precise Frequency Control Using PWM

- **Pulse Width Modulation (PWM)** technique with IC3524 allows precise control of the inverter switching frequency, ensuring that the output AC signal matches the desired frequency.
- The isolation of PWM signals using **opto-couplers** enhances system reliability and protects sensitive control units from electrical noise or disturbances.

3. Improved Waveform Quality with Multistage Filtering

- o The project addresses the common issue of square-wave output from inverters by employing an **RC multistage filter**.
- o This filtering approach effectively attenuates higher-order harmonics, resulting in a smooth **sinusoidal waveform** at the output, suitable for real-world applications requiring clean AC power.

4. Digital Monitoring and User-Friendly Interface

- o The integration of an **LCD display** controlled by an Arduino microcontroller provides real-time feedback of the output frequency, enhancing system monitoring and user interaction.

5. Cost-Effective and Versatile Solution

- o The project employs widely available components, such as MOSFETs, diodes, capacitors, and controllers, making the design **cost-effective** and suitable for implementation in low-resource environments.
- o The system can be utilized in various applications, including renewable energy systems, power backup systems, and industrial equipment requiring stable AC frequency.

6. Validation and Performance

- o The project demonstrates practical validation through oscilloscope results, where the square wave from the inverter is transformed into a clean sinusoidal wave after filtering.
- o This result proves the system's capability to maintain the desired frequency (e.g., 50/60 Hz) regardless of input variations.

5.2 Design Considerations

5.2.1 Considerations to public health and safety

- **Electrical Safety:** The system incorporates isolated PWM control signals using optocouplers to avoid electrical shocks or damages caused by high-voltage switching components.
- **Smooth and Stable Output:** The filtering unit produces a clean sinusoidal AC signal, reducing harmonic distortion, which ensures safe operation for connected devices and avoids overheating or malfunctioning.
- **Component Protection:** Proper circuit design, including protection diodes and capacitors, ensures the system operates without risks of short circuits or overvoltage, minimizing hazards.

5.2.2 Considerations to environment

- **Efficient Power Conversion:** The system minimizes energy losses during AC-DC and DC-AC conversion processes, contributing to energy efficiency.
- **Reduced Harmonics:** By filtering higher-order harmonics, the project reduces power

quality issues, which can otherwise cause excessive power consumption and strain on the grid.

- **Sustainable Component Usage:** The system uses widely available, energy-efficient components such as MOSFETs and capacitors, which have longer lifespans and minimize e-waste generation.

5.2.3 Considerations to cultural and societal needs

- **Access to Stable Power:** By stabilizing variable frequency AC, the system ensures a consistent power supply for households, industries, and rural areas where grid power is unstable.

- **Cost-Effective Design:** The project utilizes affordable and readily available components, making it suitable for low-resource settings and developing regions.

- **Improved Quality of Life:** Reliable power ensures uninterrupted operation of essential systems such as medical equipment, communication devices, and domestic appliances, contributing to societal well-being.

- **Adaptability:** The system can be customized to meet the power requirements of different regions and applications, promoting inclusivity across various cultures and communities. the project demonstrates a thoughtful design, addressing safety, environmental, and societal aspects to deliver an efficient and sustainable solution.

5.3 Limitations of Tools

- **Arduino Uno:**
 - Limited processing power
 - Limited memory
 - Analog-to-Digital Converter (ADC) resolution limitations
- **Potentiometers:**
 - Prone to wear and tear
 - Manual adjustment can be imprecise
 - Difficult to automate adjustments
- **LCD Display:**
 - Limited display size and resolution
 - Relatively slow refresh rates
- **Passive Components:**
 - Component tolerances can affect circuit performance
 - Aging can lead to performance degradation

5.4 Impact Assessment

5.4.1 Assessment of Societal and Cultural Issues

- **Access to Stable Power:** The frequency stabilizer provides a consistent and reliable power supply, benefiting communities in remote or underdeveloped areas where grid frequency instability is common.
- **Improved Quality of Life:** A stable power source ensures uninterrupted operation of critical systems such as healthcare devices, communication tools, and domestic appliances, contributing to societal well-being.
- **Adaptability:** The cost-effective design can be tailored for different regions and cultural contexts, ensuring inclusivity and usability in diverse environments.
- **Skill Development:** The project promotes learning and technological awareness, helping communities adopt innovative solutions to energy problems.

5.4.2 Assessment of Health and Safety Issues

- **Electrical Safety:** Proper isolation using optocouplers protects users from electrical hazards and accidental shocks during operation.
- **Reduced Harmonic Distortions:** The filtering unit removes harmonic distortions, preventing overheating of electrical systems, thereby improving safety for connected devices.
- **Reliability of Output:** By delivering a clean sinusoidal AC waveform, the system minimizes risks associated with malfunctioning electrical devices that could impact health and safety.
- **Safe Design Practices:** The inclusion of components such as protection diodes, capacitors, and MOSFETs ensures safe and stable operation, avoiding short circuits or overloading.

5.4.3 Assessment of Legal Issues

- **Compliance with Power Quality Standards:** The system adheres to international standards for frequency stability and waveform quality (e.g., IEEE standards), ensuring compatibility with electrical grids and devices.
- **Electrical Safety Regulations:** The project design follows electrical safety codes for isolation, grounding, and protection, ensuring it meets legal requirements for safe use.
- **Intellectual Property:** The project design and implementation can be safeguarded under patent laws, ensuring legal recognition of the innovation.

5.5 Sustainability and Environmental Impact Evaluation

- **Energy Efficiency:** The system is designed to minimize energy losses during AC-DC and DC-AC conversion, promoting efficient energy usage.
- **Reduction of Harmonic Disturbances:** By filtering higher-order harmonics, the system enhances power quality, reducing strain on electrical grids and minimizing unnecessary energy consumption.
- **Sustainable Components:** Use of long-lasting and recyclable components (e.g., MOSFETs, capacitors) reduces electronic waste and promotes environmental responsibility.

5.6 Ethical Issues

- **Adherence to Safety Standards:** The design complies with safety regulations and power quality standards to ensure no harm to users or equipment.
- **Environmental Responsibility:** The project avoids the use of hazardous or non-recyclable materials, contributing to environmental sustainability.
- **Fair Use of Resources:** Components were sourced ethically, avoiding counterfeit products and ensuring transparency with suppliers.
- **Intellectual Property:** The team adhered to ethical guidelines for using pre-existing technologies, citing references appropriately and ensuring no plagiarism in the design.
- **Ethical Challenges:** Potential challenges included selecting cost-effective materials without compromising quality and ensuring accessibility to low-resource regions. These were mitigated by choosing widely available, affordable, and efficient components.

6 Reflection on Individual and Team work

6.1 Individual Contribution of Each Member

Student ID	Name	Contribution
2006044	Mahadi Jaman	Establishing the theoretical basis of the project, implementing hardware, testing and debugging for different cases.
2006045	Shamiul Islam Anik	Implementing hardware, buying and collecting components, testing and debugging for different cases.
2006051	Ahamad Abtahi	Implementing hardware, buying and collecting components, testing and debugging for different cases, preparing presentation slides.

2006052	Md Nazmus Sakib Tushar	Implementing hardware, buying and collecting components, making simulation, testing and debugging for different cases.
2006053	Md Maksudur Rahman Turzo	Establishing the theoretical basis of the project, making simulation, implementing hardware, buying and collecting components, testing and debugging for different cases.
2006054	Joyonta Debnath	Implementing hardware, buying and collecting components, testing and debugging for different cases, insisting and contributing on perfection of the project.
2006064	Anas Rohan	Making simulation, implementing hardware, maintaining theoretical consistency, buying and collecting components, collecting necessary resources, testing and debugging for different cases.

7 Communication

Regular Team Meetings: We organized frequent team meetings, both in-person and virtual, to track project progress, provide updates on individual tasks, and address any challenges. These meetings fostered open dialogue, ensuring alignment with project objectives across all team members.

Transparency: A cornerstone of our communication strategy was transparency. We ensured all team members had access to vital project information by sharing project documentation, progress reports, and timelines. This openness fostered accountability and informed decision making.

Problem-Solving: When challenges arose, our communication strategy emphasized a collaborative, problem-solving approach. Team members actively participated in brainstorming sessions, offering diverse perspectives and solutions to overcome obstacles.

7.1 Executive Summary

This report presents the design and implementation of a Frequency Stabilizer that converts variable-frequency AC power into a fixed-frequency AC output. The system addresses the need for stable power supply frequency in scenarios where input frequency fluctuations occur, such as in renewable energy systems or unstable power grids.

The stabilizer operates through a two-stage process: AC-DC conversion and DC-AC inversion. The variable-frequency AC input is first rectified and smoothed into DC, which is then converted back into AC at a constant frequency using an inverter controlled by a PWM-based switching mechanism. An LC filter ensures that the output AC waveform is sinusoidal with minimal harmonic distortion.

Key investigations include:

- Experimental validation of the stabilizer's ability to maintain a consistent output frequency regardless of input frequency variations.
- Analysis of waveform quality, efficiency, and Total Harmonic Distortion (THD).

The results confirm that the stabilizer achieves a fixed-frequency output with high accuracy, low distortion, and efficient energy conversion. This project demonstrates the practical viability of frequency stabilization systems for ensuring power quality in variable-frequency environments.

7.2 User Manual

This project takes an unstable AC frequency input and outputs a stable AC frequency signal. Below are the steps to set up and operate the project.

Powering Up the Project To power the project, two DC sources are required:

- First DC Source: Connect this across the optocoupler.
- Second DC Source: Connect this to the SG3524 IC.
- Set the SG3524 IC DC source to 10 volts.
- Set the Optocoupler DC source to 5-6 volts.

Setting Duty Cycle

- Once the DC sources are connected, check SG3524 pin **12** to ensure the power is properly supplied and observe the duty cycle.
- Adjust the **potentiometer 1** that controls the duty cycle.
- Set the duty cycle to **50%** by adjusting the potentiometer.

Input and Output Frequency

- Provide the unstable frequency AC input at the rectifier input.
- Verify the output on the display and also use an oscilloscope at the filter output.
- Check the frequency at the final output.
- If the output frequency is not the desired frequency, use the frequency-changing **potentiometer 2** to adjust it to the desired frequency.

By following these steps, you can ensure your project delivers a stable output frequency from an unstable input.

8 Project Management and Cost Analysis

8.1 Bill of Materials

Components	Quantity	Price (Tk)
Breadboard	4	520
IRF 250 (MOSFET)	4	480
Diode	8	40
IC 3524	2	50
Optocoupler	3	105
16 x 8 LCD display	1	190
I2C display module	1	70
100k potentiometer	2	60
Arduino Uno	1	600
Capacitors, Resistors		200
Jumper Wires		200
Total		2515

9 Future Work

To enhance the Frequency Stabilizer, future work could focus on:

1. **Renewable Energy Integration:** Adapting the system for wind and solar power applications.
2. **Advanced Control Techniques:** Implementing algorithms like PID or machine learning for better efficiency.
3. **Three-Phase Support:** Extending the design for industrial three-phase systems.
4. **IoT Integration:** Adding real-time monitoring for remote diagnostics and maintenance.
5. **Efficiency Improvements:** Using advanced materials (e.g., SiC or GaN) for reduced losses and better thermal management.
6. **Portable and Scalable Design:** Creating compact and modular versions for portable or distributed power systems.

These upgrades will enhance the system's performance, scalability, and versatility in modern applications.

10 References

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